

The MSA Microanalysis Parameter Commons

MSA-branded open-source tools and a preserved knowledge base for choosing high-fidelity SEM, EDS, EBSD, and XRF acquisition parameters

MSA Strategic Initiatives Program, 2027 · Request: \$20,000 over two years (Year 1: \$9,000; Year 2: \$11,000)

Project leader

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Key personnel

- **Ronnie Guymon**, undergraduate researcher, Department of Mechanical Engineering, Brigham Young University: student lead for the curation corps and validation benchmark · [\[email\]](#)
- **Mike Standing**, BYU Electron Microscopy Facility: expert review of SEM, EDS, and EBSD records; benchmark instrument sessions · [\[email\]](#)
- **Prof. Ashley Spear**, Department of Mechanical Engineering, University of Utah: laser powder bed fusion specimens for the benchmark · [\[email\]](#)
- **Prof. Devin Rappleye**, Department of Chemical Engineering, Brigham Young University: ICP-MS ground-truth composition · [\[email\]](#)
- **Kevin Rey**, Department of Geological Sciences, Brigham Young University: WDXRF ground truth and XRF records · [\[email\]](#)

Executive summary

Quantitative SEM, EDS, EBSD, and XRF results depend on acquisition parameters, yet the reasoning behind "recommended" settings is scattered across decades of literature, vendor software, and the memories of senior microscopists. We propose the MSA Microanalysis Parameter Commons: an MSA-branded, open-source release of CALIBER, a literature-grounded, uncertainty-aware recommender for acquisition parameters, paired with a curated, versioned, DOI-citable knowledge base that records *why* each setting is chosen and links to MSA's own archives. An open, ground-truthed benchmark on additively manufactured aluminum alloys will fill a documented gap, since no controlled acquisition-parameter study exists for laser powder bed fusion AlSi10Mg. Undergraduate MSA student members will curate records under expert review, and results will reach members and adjacent communities through open webinars, an M&M tutorial, and *Microscopy Today*. Year 1 delivers the open-source release, 50 reviewed records, and the first benchmark campaign; Year 2 delivers version 1.0, 150 or more records, the public benchmark dataset, and a governance handoff to MSA's EM Data Analysis and Management Focused Interest Group. The initiative addresses Areas 2 and 6 directly and serves MSA's mission of fostering research, innovation, advancement, and promotion of microscopy.

Strategic Area Being Addressed and Justification

This initiative targets **Area 2** (MSA-branded development and dissemination of open-source tools for microscopy, microanalysis, and data management) and **Area 6** (collection, preservation, and dissemination of technical procedures, basic theory, and MSA archived material). It also involves students (Area 1), adds a recurring member benefit (Area 3), and builds academic partnerships (Area 5).

Parameters decide fidelity. Standards-based SEM-EDS can reach about $\pm 5\%$ relative uncertainty, while standardless analysis can err by $\pm 30\%$, and normalized totals hide the error [1]. Raising EBSD collection speed from 54 to 154 patterns/s cut correct indexing from 34.9% to 9.3% [2], and a 12-phase candidate list in place of the correct two yielded *zero* correct assignments despite ideal beam settings [3]. XRF has over-reported trace Mg in aluminum by 5000% relative [4]. Our own work on laser powder bed fusion (LPBF) AlSi10Mg shows the same thing: standardless EDS reported 0.94–1.64 wt% Mg, depending on beam energy, for an alloy expected near 0.5–0.7 wt%, and the fraction of reliably indexed EBSD points rose from about 40% to 77% after a beam-current correction alone.

The reasons behind settings are poorly preserved. Standards such as ISO 22309 and ASTM E1508 give operating thresholds (overvoltage ≥ 1.8 , take-off near 35°) but not the derivations or validation data needed to adapt them to a given detector, alloy, and goal; vendor-constrained algorithms left one comparison unable even to explain why two standardless systems disagreed [1]. FAIR metadata schemas record acquisition context but, by design, not the rationale for it [5], and practical expertise demonstrably leaves science when practitioners retire [6]. Open tools such as HyperSpy, kikuchipy, EMsoft, and napari analyze data after acquisition; none recommends settings, cites its sources, or reports its own uncertainty [7,8]. And much of the source knowledge is already MSA's: the strongest parameter studies and ecosystem papers sit in *Microscopy and Microanalysis* and the M&M proceedings [3,8,9], where they are hard to find and act on.

Details of the Proposal

O1 — MSA-branded open-source CALIBER. A permissively licensed release of the recommender, the uncertainty-aware feedback loop that updates recommendations as measurements arrive, and corpus-building tools, with an install-free web demo. It interoperates with HyperSpy and kikuchipy data models and NeXus metadata [5] rather than replacing them. Branding and hosting are agreed with Council at the start.

O2 — The Parameter Commons. A public, versioned knowledge base of rationale records (per modality $\times$ parameter $\times$ material class: recommended range, governing physics, primary sources, known failure modes), released with DOIs, with a contribution and review workflow and links into M&M proceedings and *Microscopy Today*. Seed parameters: EDS accelerating voltage, acquisition time (total counts), and beam current; EBSD beam current, voltage, and phase list; a baseline SEM imaging set.

O3 — Open validation benchmark. A pre-registered, ground-truthed parameter study on LPBF Al-Si specimens: 5/10/15/20 kV voltage series, beam-current and dead-time mapping, and count-level stopping rules for EDS; beam current and phase-list tests for EBSD; ICP-MS and WDXRF as independent composition references. Raw spectra and patterns are archived and re-analyzed outside vendor software, so recommended and naive settings can be compared by anyone.

O4 — Students, training, and promotion. A corps of 8 undergraduate MSA student members curates records with bounded tasks, mentored review, and visible credit on each DOI'd release [10]. Two open webinars through MSA's online channels (open to non-members), course-ready tutorial notebooks, an M&M tutorial, and a *Microscopy Today* article carry the work to members and to the additive manufacturing, materials, and geoscience communities that use these instruments daily but rarely engage with MSA.

	Year 1 goals (basis for Year 2 continuation)	Year 2 goals
Q1–Q2	Branding and hosting agreement; record schema; CALIBER v0.1 release; 50 reviewed seed records; 4 curators recruited	Benchmark dataset released with DOI and FAIR metadata; contribution workflow opens to members; CALIBER v1.0 with XRF coverage
Q3–Q4	First benchmark campaign run and analyzed; public beta; first webinar; M&M 2028 tutorial proposal; <i>Microscopy Today</i> draft	150+ records; second webinar; M&M 2028 tutorial; governance handoff; final report to Council

Measuring success. We will report against fixed targets: published records (50 in Year 1, 150 in Year 2); records linked to MSA archived material (25, 75); student curators trained (4, 8); webinar attendance, including the non-member share (target 40% non-members); web-demo sessions and repository downloads; benchmark dataset downloads; and, from an opt-in survey at demo sign-up and webinar registration, how many users are MSA members and how many joined after using the resource. Year 1 succeeds if v0.1, 50 reviewed records, and the first campaign are public; the initiative succeeds if the Commons is self-sustaining under MSA governance at the end of Year 2.

Innovation and Impact

Innovation. CALIBER is the first open tool that works *before* acquisition: it retrieves the literature, applies physics models, and states its uncertainty for a given sample and goal, then updates as measurements arrive. Existing autonomous-microscopy work is task-specific and policy-driven rather than literature-grounded [7]. The Commons is also new in kind: protocol repositories keep the procedural *what*, while the Commons keeps the expert *why* (alternatives considered, failure modes, stopping criteria), each claim tied to a primary source and, where possible, to an open benchmark result. The benchmark is pre-registered, so the recommendations can be shown wrong; that risk is deliberate and is what makes the records trustworthy.

Impact on MSA's strategic goals. The initiative serves each part of MSA's mission. It *fosters research* with benchmark data the literature lacks and with more defensible measurements in every lab that adopts validated settings. It is an *innovation* in open microscopy infrastructure that connects the existing ecosystem instead of duplicating it. It *advances* practice by turning retirement-vulnerable expertise into citable, versioned records [6]. And it *promotes microscopy* by carrying MSA's name, and the case for rigorous microanalysis, to students and adjacent communities through tools whose every suggestion cites the microscopy literature. Members gain a recurring benefit: faster, more reliable data on unfamiliar samples. MSA becomes the recognized home of preserved, citable microanalysis practice, as rOpenSci and MolSSI became for their communities [10,11].

Plan for Long-Term Support

This activity does not require financial support from MSA after the end of the Strategic Initiative funding. In Year 2, governance passes to MSA's EM Data Analysis and Management Focused Interest Group under a lightweight editor-plus-reviewers model, with a liaison to the Microanalysis Society's microanalytical-standards group (FIGMAS) for EDS records. This follows the precedent literature: society-branded resources survive when a trusted organization governs them, they are integrated with existing community activities (here, M&M and *Microscopy Today*), and contributors receive credit, and they fail when they rest on volunteers alone, detached from established infrastructure [10,11]. Ongoing costs are small (a static site, a public repository, and free DOI archiving), and the Vertical Cloud Lab will cover hosting and model-inference costs from its own funds. Curation continues through the BYU student pipeline and member contributions credited on each release.

References. [1] V. Tong, K. Mingard, NPL Report MAT 135 (2026). [2] S. I. Wright et al., *Ultra-microscopy* 159, 81 (2015). [3] K. Kaufmann, K. S. Vecchio, *Microsc. Microanal.* 27, 776 (2021). [4] P. Seidel et al., *Metals* 11, 736 (2021). [5] J. A. Taillon et al., *MRS Bull.* 50, 793 (2025). [6] P. F. Rainford et al., *Nat. Commun.* (2026). [7] S. V. Kalinin et al., *ACS Nano* 15, 12604 (2021). [8] J. Wei et al., *Microsc. Microanal.* 28 (2022). [9] M. Kühbach et al., *Microsc. Microanal.* 28 (2022). [10] D. S. Katz et al., *Comput. Sci. Eng.* 21, 8 (2019). [11] C. M. Schweik, in *Understanding Knowledge as a Commons* (MIT Press, 2007).

Budget Justification

Line item	Year 1	Year 2	Total
Student curator stipends	\$4,000	\$4,000	\$8,000
Instrument time for the open benchmark	\$3,500	\$1,000	\$4,500
Compute and hosting	\$1,000	\$1,000	\$2,000
Webinars and tutorial notebooks	\$500	\$500	\$1,000
M&M 2028 tutorial: student curator travel	—	\$2,500	\$2,500
Open-access publication of the benchmark	—	\$2,000	\$2,000
Total requested	\$9,000	\$11,000	\$20,000

- **Student curator stipends (\$8,000).** Eight undergraduate MSA student members (four per year) at \$1,000 each, about 60 hours of mentored curation apiece: extracting sources, drafting records, and revising them after expert review. These are training stipends that bring students into quantitative microanalysis (Area 1); they replace no volunteer effort. Expert review by members and all faculty effort remain volunteer contributions.
- **Instrument time (\$4,500).** SEM, EDS, and EBSD sessions at the BYU Electron Microscopy Facility for the pre-registered benchmark in Year 1, and WDXRF sessions for XRF coverage in Year 2. Every session produces data released publicly. The Vertical Cloud Lab covers the remaining instrument time, specimens, standards, and sample preparation.
- **Compute and hosting (\$2,000).** Language-model inference for the public web demo and cloud hosting for the demo and Commons site; DOI archiving through Zenodo is free.
- **Webinars and tutorial notebooks (\$1,000).** Recording, captioning, and editing for two open webinars delivered through MSA's online channels, and packaging of course-ready notebooks.
- **M&M 2028 tutorial (\$2,500).** Registration and travel for two student curators to co-present the tutorial and the benchmark results.
- **Open-access publication (\$2,000).** Article processing charges so the benchmark paper is freely readable alongside its open dataset.

No indirect costs, faculty salary, or equipment purchases are requested. Year 2 funds are requested only on demonstrating the Year 1 goals.